# SOLUTIONS: SYSTEMS IDENTIFICATION

1. Solution

a) Consider the linear estimator

$$\hat{S}(\alpha,\beta,\gamma) = \alpha S_1 + \beta S_2 + \gamma S_3.$$

Applying the expected value operator yields

$$\mathbb{E}[\hat{S}(\alpha,\beta,\gamma)] = \alpha\mathbb{E}[x(1)] + \beta\mathbb{E}[x(2)] + \gamma\mathbb{E}[x(3)] = (\alpha+\beta+\gamma)\bar{x},$$

where $\bar{x} = \mathbb{E}[x(1)] = \mathbb{E}[x(2)] = \mathbb{E}[x(3)]$. Then, the estimator $\hat{S}(\alpha,\beta,\gamma)$ is unbiased if $\alpha+\beta+\gamma = 1$.

[ 3 Marks ]

b) By applying the expected value operator to both sides of the stochastic process difference equation, one gets

$$\mathbb{E}[x(t+1)] = \frac{1}{2}\mathbb{E}[x(t)] + \mathbb{E}[w(t)].$$

Since the process is stationary and $\mathbb{E}[w(t)] = \bar{w}$, we have

$$\mathbb{E}[x(t)] = \frac{1}{2}\mathbb{E}[x(t)] + \bar{w}.$$

Solving for $\mathbb{E}[x(t)]$ yields

$$\mathbb{E}[x(t)] = 2\bar{w}.$$

To obtain the variance of $x(t)$, we first define $\tilde{x}(t) = x(t) - \mathbb{E}[x(t)] = x(t) - 2\bar{w}$ and $\tilde{w}(t) = w(t) - \mathbb{E}[w(t)] = w(t) - \bar{w}$. According to the stochastic process difference equation, $\tilde{x}(t)$ satisfies

$$\begin{aligned}\tilde{x}(t+1) = x(t+1) - 2\bar{w} &= \frac{1}{2}x(t) + w(t) - 2\bar{w} \\ &= \frac{1}{2}(\tilde{x}(t) + 2\bar{w}) + (\tilde{w}(t) + \bar{w}) - 2\bar{w} \\ &= \frac{1}{2}\tilde{x}(t) + \tilde{w}(t).\end{aligned} \tag{1.1}$$

Accordingly, since $\alpha+\beta+\gamma = 1$, we obtain the variance of $x(t)$ as follows:

$$\begin{aligned}\text{var}[x(t)] &= \mathbb{E}[\tilde{x}(t)^2] \\ &= \mathbb{E}\left[\left(\frac{1}{2}\tilde{x}(t-1) + \tilde{w}(t-1)\right)^2\right] \\ &= \frac{1}{4}\mathbb{E}[\tilde{x}(t-1)^2] + \mathbb{E}[\tilde{w}(t-1)^2] + \mathbb{E}[\tilde{x}(t-1)\tilde{w}(t-1)].\end{aligned} \tag{1.2}$$

Since there is no correlation between $\tilde{x}(t-1)$ and $\tilde{w}(t-1)$, $\mathbb{E}[\tilde{x}(t-1)\tilde{w}(t-1)] = 0$. Moreover, by stationarity, $\mathbb{E}[\tilde{x}(t-1)^2] = \mathbb{E}[\tilde{x}(t)^2] = \text{var}[x(t)]$. Thus, as $\mathbb{E}[\tilde{w}(t-1)^2] = 2$, (1.2) can be restated as follows:

$$\text{var}[x(t)] = \frac{1}{4}\text{var}[x(t)] + 2$$

Answers

which yields

$$\text{var}[x(t)] = \frac{8}{3}.$$

[ 4 Marks ]

c) By defining $\tilde{x}(i) = x(i) - \mathbb{E}[x(i)]$ for all $i = 1,2,3$, the covariance matrix of the random variable $[x(1)\ \ x(2)\ \ x(3)]$ is as follows:

$$\text{cov}[x(1)\ \ x(2)\ \ x(3)] = \begin{bmatrix} \mathbb{E}[\tilde{x}(1)^2] & \mathbb{E}[\tilde{x}(1)\tilde{x}(2)] & \mathbb{E}[\tilde{x}(1)\tilde{x}(3)] \\ \star & \mathbb{E}[\tilde{x}(2)^2] & \mathbb{E}[\tilde{x}(2)\tilde{x}(3)] \\ \star & \star & \mathbb{E}[\tilde{x}(3)^2] \end{bmatrix}.$$

Since $\mathbb{E}[\tilde{x}(i)^2] = 8/3$ for all $i = 1,2,3$, it remains to compute the quantities $\mathbb{E}[\tilde{x}(1)\tilde{x}(2)]$, $\mathbb{E}[\tilde{x}(1)\tilde{x}(3)]$, and $\mathbb{E}[\tilde{x}(2)\tilde{x}(3)]$. Now, by considering (1.1), we obtain

$$\mathbb{E}[\tilde{x}(1)\tilde{x}(2)] = \mathbb{E}\left[\tilde{x}(1)\left(\frac{1}{2}\tilde{x}(1) + \tilde{w}(1)\right)\right] = \frac{1}{2}\text{var}[x(1)] + \mathbb{E}[\tilde{x}(1)\tilde{w}(1)].$$

Since $\tilde{x}(1)$ and $\tilde{w}(1)$ are not correlated, we get $\mathbb{E}[\tilde{x}(1)\tilde{w}(1)] = 0$. Therefore, we obtain

$$\mathbb{E}[\tilde{x}(1)\tilde{x}(2)] = \frac{1}{2}\text{var}[x(1)] = \frac{4}{3}.$$

In a similar way,

$$\begin{aligned}\mathbb{E}[\tilde{x}(2)\tilde{x}(3)] &= \mathbb{E}\left[\tilde{x}(2)\left(\frac{1}{2}\tilde{x}(2) + \tilde{w}(2)\right)\right] \\ &= \frac{1}{2}\text{var}[x(2)] + \mathbb{E}[\tilde{x}(2)\tilde{w}(2)] = \frac{1}{2}\text{var}[x(2)] = \frac{4}{3}.\end{aligned}$$

Moreover,

$$\begin{aligned}\mathbb{E}[\tilde{x}(1)\tilde{x}(3)] &= \mathbb{E}\left[\tilde{x}(1)\left(\frac{1}{2}\tilde{x}(2) + \tilde{w}(2)\right)\right] \\ &= \frac{1}{2}\mathbb{E}[\tilde{x}(1)\tilde{x}(2)] + \mathbb{E}[\tilde{x}(1)\tilde{w}(2)] = \frac{1}{2}\mathbb{E}[\tilde{x}(1)\tilde{x}(2)] = \frac{2}{3}.\end{aligned}$$

[ 5 Marks ]

d) First, by letting $\bar{x} = \mathbb{E}[x(1)] = \mathbb{E}[x(2)] = \mathbb{E}[x(3)]$, we compute the variance of $\hat{S}(\alpha,\beta,\gamma)$ as follows:

$$\begin{aligned}\text{var}(\hat{S}(\alpha,\beta,\gamma)) &= \mathbb{E}\left[(\hat{S}(\alpha,\beta,\gamma) - \mathbb{E}[\hat{S}(\alpha,\beta,\gamma)])^2\right] = \mathbb{E}\left[(\alpha x(1) + \beta x(2) + \gamma x(3) - \bar{x})^2\right] \\ &= \mathbb{E}\left[(\alpha(x(1)-\bar{x}) + \beta(x(2)-\bar{x}) + \gamma(x(3)-\bar{x}))^2\right] \\ &= \mathbb{E}\left[(\alpha\tilde{x}(1) + \beta\tilde{x}(2) + \gamma\tilde{x}(3))^2\right] \\ &= \alpha^2\text{var}[\tilde{x}(1)] + \beta^2\text{var}[\tilde{x}(2)] + \gamma^2\text{var}[\tilde{x}(3)] \\ &+ 2\alpha\beta\mathbb{E}[\tilde{x}(1)\tilde{x}(2)] + 2\alpha\gamma\mathbb{E}[\tilde{x}(1)\tilde{x}(3)] + 2\beta\gamma\mathbb{E}[\tilde{x}(2)\tilde{x}(3)],\end{aligned}$$

in which we have used the fact that $\alpha + \beta + \gamma = 1$. By considering the values obtained in the answer to Question 1-c), one gets

$$\text{var}(\hat{S}(\alpha,\beta,\gamma)) = \frac{8}{3}(\alpha^2 + \beta^2 + \gamma^2) + \frac{8}{3}\alpha\beta + \frac{4}{3}\alpha\gamma + \frac{8}{3}\beta\gamma.$$

Answers

From $\alpha+\beta+\gamma=1$, it follows that $\gamma=1-\alpha-\beta$, and accordingly, $\text{var}(\hat{S}(\alpha,\beta,\gamma))$ can be obtained as follows:

$$\text{var}(\hat{S}(\alpha,\beta,\gamma)) = 4\alpha^2+\frac{8}{3}\beta^2-4\alpha-\frac{8}{3}\beta+4\alpha\beta+\frac{8}{3}. \qquad (1.3)$$

Now, to obtain $\alpha$ and $\beta$ minimizing $\text{var}(\hat{S}(\alpha,\beta,\gamma))$, we should obtain $\alpha$ and $\beta$ such that

$$\frac{\partial}{\partial\alpha}\text{var}(\hat{S}(\alpha,\beta,\gamma)) = 0$$
$$\frac{\partial}{\partial\beta}\text{var}(\hat{S}(\alpha,\beta,\gamma)) = 0,$$

which can be restated as follows:

$$2\alpha+\beta-1=0$$
$$3\alpha+4\beta-2=0$$

which yield

$$\alpha^\circ=\frac{2}{5}, \quad \beta^\circ=\frac{1}{5}.$$

Therefore, $\gamma^\circ=1-\alpha^\circ-\beta^\circ=2/5$.

[ 8 Marks ]

Answers

2. Solution

a) Based on the block diagram, one gets

$$x_1(t+1) = \frac{1}{5}x_3(t) + v(t) \tag{2.1}$$
$$x_2(t+1) = a_2 x_1(t) + v(t) \tag{2.2}$$
$$x_3(t+1) = a_1 x_1(t) + x_2(t) - \frac{1}{2}x_3(t) \tag{2.3}$$
$$y(t) = x_3(t). \tag{2.4}$$

By time-shifting Equation (2.3), we obtain

$$x_3(t+2) = a_1 x_1(t+1) + x_2(t+1) - \frac{1}{2}x_3(t+1). \tag{2.5}$$

Substituting $x_1(t+1)$ and $x_2(t+1)$ from (2.1) and (2.2) into (2.5) yields

$$x_3(t+2) = \frac{1}{5}a_1 x_3(t) + a_1 v(t) + a_2 x_1(t) + v(t) - \frac{1}{2}x_3(t+1). \tag{2.6}$$

By time-shifting Equation (2.6), we obtain

$$x_3(t+3) = \frac{1}{5}a_1 x_3(t+1) + a_1 v(t+1) + a_2 x_1(t+1) + v(t+1) - \frac{1}{2}x_3(t+2). \tag{2.7}$$

Now, substituting $x_1(t+1)$ from (2.1) into (2.7) yields

$$x_3(t+3) = \frac{1}{5}a_1 x_3(t+1) + a_1 v(t+1) + \frac{1}{5}a_2 x_3(t) + a_2 v(t) + v(t+1) - \frac{1}{2}x_3(t+2)$$

from which, since $y(t) = x_3(t)$, the following ARMA model can be obtained:

$$y(t) = -\frac{1}{2}y(t-1) + \frac{1}{5}a_1 y(t-2) + \frac{1}{5}a_2 y(t-3) + (a_1+1)v(t-2) + a_2 v(t-3). \tag{2.8}$$

[ 5 Marks ]

b) From (2.8), We obtain the transfer function $G(z) = y(z)/v(z)$ as follows:

$$G(z) = \frac{(a_1+1)z^{-2} + a_2 z^{-3}}{1 + \frac{1}{2}z^{-1} - \frac{1}{5}a_1 z^{-2} - \frac{1}{5}a_2 z^{-3}}.$$

For $a_1 = 1$ and $a_2 = 0$, the poles of $G(z)$ can be obtained by solution of the following equation:

$$z^2 + \frac{1}{2}z - \frac{1}{5} = 0$$

which are given by

$$z_1 = -\frac{1}{4} + \sqrt{\frac{1}{16} + \frac{1}{5}} = 0.2623$$
$$z_2 = -\frac{1}{4} - \sqrt{\frac{1}{16} + \frac{1}{5}} = -0.7623.$$

Since the poles are in the open unit circle, then the process $y(\cdot)$ is stationary.

[ 4 Marks ]

Answers

c) By letting $a_1 = 1$ and $a_2 = 0$, we obtain

$$\begin{aligned} x_1(t+1) &= \frac{1}{5}x_3(t) + v(t) \\ x_2(t+1) &= v(t) \\ x_3(t+1) &= x_1(t) + x_2(t) - \frac{1}{2}x_3(t) \end{aligned}$$

Thanks to the stationarity established in the answer to Question 2-b) and owing to the fact that $\mathbb{E}[v(t)] = 1$, one gets

$$\mathbb{E}[x_1(t)] = \frac{1}{5}\mathbb{E}[x_3(t)] + 1 \tag{2.9}$$

$$\mathbb{E}[x_2(t)] = 1 \tag{2.10}$$

$$\mathbb{E}[x_3(t)] = \mathbb{E}[x_1(t)] + \mathbb{E}[x_2(t)] - \frac{1}{2}\mathbb{E}[x_3(t)]. \tag{2.11}$$

Substituting (2.9) and (2.10) into (2.11) yields

$$\mathbb{E}[x_3(t)] = \frac{1}{5}\mathbb{E}[x_3(t)] + 2 - \frac{1}{2}\mathbb{E}[x_3(t)]$$

which implies that $\mathbb{E}[x_3(t)] = \frac{20}{13}$. Now, substituting $\mathbb{E}[x_3(t)] = \frac{20}{13}$ into (2.9) yields $\mathbb{E}[x_1(t)] = \frac{17}{13}$ and $\mathbb{E}[x_2(t)] = 1$.

[ 4 Marks ]

d) We need to obtain $\gamma_y(0)$, $\gamma_y(1)$, and $\gamma_y(2)$. According to the answer to Question 2-b), for $a_1 = 1, a_2 = 0$, the stochastic process $y(\cdot)$ is stationary. By multiplying both sides of (2.8) by $y(t)$, one obtains

$$\begin{aligned} \gamma_y(0) = \mathbb{E}[y(t)^2] &= \mathbb{E}\left[-\frac{1}{2}y(t)y(t-1) + \frac{1}{5}y(t)y(t-2) + 2y(t)v(t-2)\right] \\ &= -\frac{1}{2}\gamma_y(1) + \frac{1}{5}\gamma_y(2) + 2\mathbb{E}[y(t)v(t-2)]. \end{aligned} \tag{2.12}$$

By multiplying both sides of (2.8) by $v(t-2)$ and taking the expectation, one gets

$$\begin{aligned} \mathbb{E}[y(t)v(t-2)] &= \mathbb{E}\left[\left(-\frac{1}{2}y(t-1) + \frac{1}{5}y(t-2) + 2v(t-2)\right)v(t-2)\right] \\ &= \mathbb{E}\left[2v(t-2)^2\right] = 2\gamma_v(0) \end{aligned} \tag{2.13}$$

in which we have used the fact that $v(t-2)$ is uncorrelated with $y(t-1)$ and $y(t-2)$. Since $\gamma_v(0) = \text{var}[v(t)] + \mathbb{E}[v(t)]^2 = 1 + 1 = 2$, then from (2.12) and (2.13), it follows that

$$\gamma_y(0) = -\frac{1}{2}\gamma_y(1) + \frac{1}{5}\gamma_y(2) + 8. \tag{2.14}$$

Answers

In the same way, we obtain

$$\begin{aligned}\gamma_y(1) &= \mathbb{E}[y(t)y(t-1)] \\ &= \mathbb{E}\left[\left(-\frac{1}{2}y(t-1)+\frac{1}{5}y(t-2)+2v(t-2)\right)y(t-1)\right] \qquad (2.15)\\ &= -\frac{1}{2}\gamma_y(0)+\frac{1}{5}\gamma_y(1), \\ \gamma_y(2) &= \mathbb{E}[y(t)y(t-2)] \\ &= \mathbb{E}\left[\left(-\frac{1}{2}y(t-1)+\frac{1}{5}y(t-2)+2v(t-2)\right)y(t-2)\right] \qquad (2.16)\\ &= -\frac{1}{2}\gamma_y(1)+\frac{1}{5}\gamma_y(0)\end{aligned}$$

in which we have used the fact that $v(t-2)$ is uncorrelated with $y(t-2)$ and

$$\mathbb{E}[y(t-1)v(t-2)] = \mathbb{E}\left[\left(-\frac{1}{2}y(t-2)+\frac{1}{5}y(t-3)+2v(t-3)\right)v(t-2)\right] = 0.$$

Now, according to (2.14), (2.15), and (2.16), one obtains $\gamma_y(0) = 13.6752$, $\gamma_y(1) = -8.5470$, and $\gamma_y(2) = 7.0085$.

[ 7 Marks ]

Answers

3. Solution

a) By substituting the expression of $C(t)$ into the model, one obtains

$$S(t+1) = (1-\alpha)S(t) - \beta(S(t) - S(t-1))^3 + (1-\gamma)U(t) + I(t), \quad t \geq 1. \tag{3.1}$$

[ 3 Marks ]

b) By considering

$$\theta_1 = 1-\alpha, \theta_2 = -\beta, \theta_3 = 1-\gamma,$$

from (3.1), one gets

$$S(t+1) = \sum_{i=1}^{N} \theta_i p_i(t, t-1) + I(t), \quad t \geq 1$$

where $N = 3$ and

$$p_1(t,t-1) = S(t), p_2(t,t-1) = (S(t) - S(t-1))^3, p_3(t,t-1) = U(t).$$

[ 3 Marks ]

c) According to the parametric model obtained in the answer to Question 3-b), the estimation $\hat{S}(t+1, \hat{\theta})$ of $S(t+1)$ can be written as follows:

$$\hat{S}(t+1, \hat{\theta}) = p(t, t-1)^\top \hat{\theta}$$

where the following definitions hold:

$$p(t,t-1) = \begin{bmatrix} S(t) \\ (S(t) - S(t-1))^3 \\ U(t) \end{bmatrix},$$

$$\hat{\theta} = \begin{bmatrix} \hat{\theta}_1 & \hat{\theta}_2 & \hat{\theta}_3 \end{bmatrix}^\top.$$

To estimate $\theta$, we consider the following estimation error:

$$e(j) = S(j+1) - \hat{S}(t+1, \hat{\theta}) = S(j+1) - p(j, j-1)^\top \hat{\theta}, j = 1, 2, \ldots, M-1,$$

and we consider a cost function of the form:

$$J(\hat{\theta}) = \frac{1}{M-1} \sum_{j=1}^{M-1} e(j)^2 = \frac{1}{M-1} \sum_{j=1}^{M-1} \left(S(j+1) - p(j, j-1)^\top \hat{\theta}\right)^2.$$

The objective is to obtain the estimated values $\hat{\theta} = \begin{bmatrix} \hat{\theta}_1 & \hat{\theta}_2 & \hat{\theta}_3 \end{bmatrix}^\top$ subject to $\min_{\hat{\theta}} J(\hat{\theta})$. Therefore, it is sufficient to find $\hat{\theta}$ making $\frac{\partial J(\hat{\theta})}{\partial \hat{\theta}} = [0\ 0\ 0]^\top$. We have

$$\begin{aligned} \frac{\partial J(\hat{\theta})}{\partial \hat{\theta}} &= \frac{2}{M-1} \sum_{j=1}^{M-1} \left(S(j+1) - p(j, j-1)^\top \hat{\theta}\right) \frac{\partial}{\partial \hat{\theta}} \left(S(j+1) - p(j, j-1)^\top \hat{\theta}\right) \\ &= -\frac{2}{M-1} \sum_{j=1}^{M-1} p(j, j-1) \left(S(j+1) - p(j, j-1)^\top \hat{\theta}\right). \end{aligned}$$

Now, to make $\frac{\partial J(\hat{\theta})}{\partial \hat{\theta}} = [0\ 0\ 0]^\top$, we should have

$$\sum_{j=1}^{M-1} p(j, j-1) p(j, j-1)^\top \hat{\theta} = \sum_{j=1}^{M-1} S(j+1) p(j, j-1).$$

Answers

Therefore, if the matrix $\sum_{j=1}^{M-1} p(j,j-1)p(j,j-1)^\top$ is full rank, $\hat{\theta}$ can be obtained as follows:

$$\hat{\theta} = \left( \sum_{j=1}^{M-1} p(j,j-1)p(j,j-1)^\top \right)^{-1} \sum_{j=1}^{M-1} S(j+1)p(j,j-1).$$

[ 6 Marks ]

d) For each of the samples given, we obtain $\sum_{j=1}^{M-1} p(j,j-1)p(j,j-1)^\top$ as follows:

i) For the first set

$$G(2) = \sum_{j=1}^{2} p(j,j-1)p(j,j-1)^\top = \begin{bmatrix} 5 & 2 & 6 \\ 2 & 1 & 2 \\ 6 & 2 & 8 \end{bmatrix}.$$

ii) For the second set

$$G(6) = \sum_{j=1}^{6} p(j,j-1)p(j,j-1)^\top = \begin{bmatrix} 56.5 & 7.25 & 45.5 \\ 7.25 & 2.0469 & 6 \\ 45.5 & 6 & 39 \end{bmatrix}.$$

Since the first matrix, $G(2)$, is singular, the procedure does not lead to a unique solution. This is due to the fact the number of samples used is smaller than the number of parameters. Instead, the other matrix $G(6)$ is nonsingular, and thus it leads to a unique solution.

[ 5 Marks ]

e) According to the answer to Question 3-d), the set of samples i) does not lead to a unique solution and hence the computation of $\text{var}[\hat{\theta}_1]$ is pointless. For the case ii), we have $\text{var}(\hat{\theta}_1) = \sigma_I^2 b_{11}$, where $b_{11}$ is the first entry of the inverse of the matrix $G(6)$ and equals 0.2975. Is is worth noting that adding more samples than the ones considered in case ii) would give smaller values of $\text{var}[\hat{\theta}_1]$.

[ 3 Marks ]

Answers

4. Solution

a) The predictor corresponding to the family of the models $\mathscr{M}(\theta)$ is as follows:

$$\hat{\mathscr{M}}(\theta): \quad \hat{v}(t|t-1) = a_1 v(t-1) + a_2 v(t-2).$$

The estimate $\hat{\theta}(N) = [\hat{a}_1(N), \hat{a}_2(N)]$ almost surely converges to the minimum of

$$\begin{aligned}\bar{J}(a_1,a_2) =& \mathbb{E}\left[(v(t)-\hat{v}(t|t-1))^2\right] = \mathbb{E}\Bigg[\Big(e(t)-\frac{1}{2}e(t-1)-\frac{1}{2}e(t-2)\\ &-a_1 e(t-1)+\frac{1}{2}a_1 e(t-2)+\frac{1}{2}a_1 e(t-3)\\ &-a_2 e(t-2)+\frac{1}{2}a_2 e(t-3)+\frac{1}{2}a_2 e(t-4)\Big)^2\Bigg]\\ =& \left(1+\left(a_1+\frac{1}{2}\right)^2+\left(a_2+\frac{1}{2}-\frac{a_1}{2}\right)^2+\frac{1}{4}(a_1+a_2)^2+\frac{1}{4}a_2^2\right)\mathrm{var}(e).\end{aligned}$$

Therefore, to obtain the minimum of $\bar{J}(a_1,a_2)$, we should obtain $a_1$ and $a_2$ such that

$$\begin{aligned}\frac{\partial}{\partial a_1}\bar{J}(a_1,a_2) &= 0\\ \frac{\partial}{\partial a_2}\bar{J}(a_1,a_2) &= 0\end{aligned}$$

implying that

$$\begin{aligned}3a_1 - \frac{1}{2}a_2 + \frac{1}{2} &= 0\\ -\frac{1}{2}a_1 + 3a_2 + 1 &= 0\end{aligned}$$

which have the solutions for $a_1 = -0.2286$ and $a_2 = -0.3714$ and hence $\bar{\theta}_1 = -0.2286$ and $\bar{\theta}_2 = -0.3714$.

[ 4 Marks ]

b) We have

$$G(z) = \frac{C(z)}{A(z)} = \frac{1}{1-a_1 z^{-1}-a_2 z^{-2}} = 1 + z^{-1}\frac{a_1+a_2 z^{-1}}{1-a_1 z^{-1}-a_2 z^{-2}}$$

Therefore, since $v(t) = A(z)/C(z)y(t)$, the optimal one-step ahead predictor can be stated as follows:

$$\begin{aligned}\hat{v}(t+1|t) &= \frac{a_1+a_2 z^{-1}}{1-a_1 z^{-1}-a_2 z^{-2}}G(z)^{-1} = a_1 v(t) + a_2 v(t-1)\\ &= -0.2286 v(t) - 0.3714 v(t-1).\end{aligned}$$

[ 4 Marks ]

c) We have

$$G(z) = \frac{C(z)}{A(z)} = \frac{1}{1-a_1 z^{-1}-a_2 z^{-2}} = 1 + z^{-1}\frac{a_1+a_2 z^{-1}}{1-a_1 z^{-1}-a_2 z^{-2}}$$

Answers

and then, in the next step one gets

$$G(z) = \frac{C(z)}{A(z)} = 1 + a_1 z^{-1} + z^{-2} \frac{a_2 + a_1^2 + a_1 a_2 z^{-1}}{1 - a_1 z^{-1} - a_2 z^{-2}}.$$

Therefore, since $v(t) = A(z)/C(z)y(t)$, the optimal two-steps ahead predictor can be stated as follows:

$$\begin{aligned}\hat{v}(t+2|t) &= \frac{a_2 + a_1^2 + a_1 a_2 z^{-1}}{1 - a_1 z^{-1} - a_2 z^{-2}} G(z)^{-1} = (a_2 + a_1^2) v(t) + a_1 a_2 v(t-1) \\ &= -0.3191 v(t) + 0.0849 v(t-1).\end{aligned}$$

[ 6 Marks ]

d) The variance of the prediction error $e_1(t) = v(t+1) - \hat{v}(t+1|t)$ can be expressed as follows:

$$\begin{aligned}&\mathbb{E}\left[(v(t+1) - \hat{v}(t+1|t))^2\right] = \mathbb{E}\left[(v(t+1) - a_1 v(t) - a_2 v(t-1))^2\right] \\ &= \mathbb{E}\left[\left(e(t+1) - \frac{1}{2}e(t) - \frac{1}{2}e(t-1) - a_1 e(t) + a_1 \frac{1}{2}e(t-1) + a_1 \frac{1}{2}e(t-2)\right.\right. \\ &\left.\left. - a_2 e(t-1) + a_2 \frac{1}{2}e(t-2) + a_2 \frac{1}{2}e(t-3)\right)^2\right] \\ &= \left[1 + \left(\frac{1}{2} + a_1\right)^2 + \left(-\frac{1}{2} + \frac{a_1}{2} - a_2\right)^2 + \left(\frac{a_1}{2} + \frac{a_2}{2}\right)^2 + \frac{a_2^2}{4}\right] \mathrm{var}(e) = 1.2571.\end{aligned}$$

Moreover, the variance of the prediction error $e_2(t) = v(t+2) - \hat{v}(t+2|t)$ can be expressed as follows:

$$\begin{aligned}&\mathbb{E}\left[(v(t+2) - \hat{v}(t+2|t))^2\right] \\ &= \mathbb{E}\left[(v(t+2) - (a_2 + a_1^2) v(t) - a_1 a_2 v(t-1))^2\right] \\ &= \mathbb{E}\left[\left(e(t+2) - \frac{1}{2}e(t+1) - \frac{1}{2}e(t) - (a_2 + a_1^2)\left(e(t) - \frac{1}{2}e(t-1) - \frac{1}{2}e(t-2)\right)\right.\right. \\ &\quad \left.\left. - a_1 a_2 \left(e(t-1) - \frac{1}{2}e(t-2) - \frac{1}{2}e(t-3)\right)\right)^2\right] \\ &= \left(1 + \frac{1}{4} + \left(\frac{1}{2} + a_2 + a_1^2\right)^2 + \left(\frac{a_2 + a_1^2}{2} - a_1 a_2\right)^2 + \left(\frac{a_2 + a_1^2}{2} + \frac{a_1 a_2}{2}\right)^2 + \frac{a_1^2 a_2^2}{4}\right) \mathrm{var}[e(t)] \\ &= 1.3580.\end{aligned}$$

Accordingly, the variance of the error when the optimal one-step ahead predictor is used is less than when the optimal two-steps ahead predictor is used. This is due to the fact that in the one-step ahead predictor, $\hat{v}(t+1|t)$ is a function of samples closer in time to the predicted quantity than those used by the two-steps ahead predictor.

[ 6 Marks ]

Answers